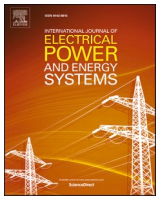
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Overcurrent protection coordination with flexible partitioning of active distribution systems into multiple microgrids

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ABSTRACT

Recently, sectionalizing active distribution systems (ADS) into partitioned microgrids has been proposed to enhance the reliability and supply-security but can have adverse effects on the performance of the protection scheme. The previous protection schemes based on overcurrent relays have been proposed for active distribution systems (ADS) with no partitioning capability. This paper analyzes the impact of microgrid partitioning on the performance of the microgrid protection schemes relying on the standard inverse time overcurrent relays (ITOCRs) and introduces a protection scheme that relies on inverse time admittance relay (ITAR) for ADS with partitioning capability. The introduced protection scheme is tested on a modified PG&E 69-bus equipped with distributed generation and partitioned into several microgrids. The protection coordination problem is formulated as a constrained nonlinear programming (NLP) problem considering grid-connected, islanded, as well as partitioned operation modes. Simulation results demonstrate that the ITAR characteristic shows promising performance in comparison to the standard ITOCR characteristic achieving minimal relay operating time and coordination failure.

1. Introduction

Islanded partition of active distribution system (ADS) into multiple microgrids can enhance the control and operation of the distribution network, as reported in IEEE Std 1547.4-2011 [1]. Accordingly, different research methods have been conducted on sectionalizing the ADS into partitioned microgrids [2–6]. In these methods, different aspects have been considered when the ADS is required to be partitioned. For example, the ADS can be partitioned based on optimizing the supply adequacy [3] or improving the reliability and supply security [4]. In [4,5], the ADS can be clustered based on minimizing the energy not supplied in grid-connected and islanded operation modes and minimizing the exchange of power between partitioned microgrids and the imbalance of generation-load within each microgrid, respectively. In [6], the ADS partitioning is achieved considering communication and control requirements.

Over the past few decades, several protection schemes have been introduced to obtain the optimal relay coordination based on the tripping characteristic of the standard inverse time overcurrent relay (ITOCR), such as in [7,8]. However, these schemes do not consider the changes in the network configuration or the effect of ADS partitioning, which in turn may lead to incorrect operation of the relays and violation of their time coordination [9].

On the other hand, other protection schemes take into account the changes in the network configuration [10–17]. In [10], an iterative technique has been introduced for the protection coordination of directional overcurrent relays of a meshed system, taking into consideration the changes in the system topology due to transient configurations, when only one of the main relays operates. However, only time-dial setting is only the decision variable of the optimization problem, and thus, a linear programming algorithm is utilized to obtain the value of the time-dial setting for each relay. A hybrid genetic algorithm (GA) and interval linear programming (ILP) algorithm are proposed to solve the protection coordination problem for a meshed distribution system considering multiple network configurations in [11,12], respectively. For both algorithms, multiple network configurations resulting from a single line outage are considered by combining all constraints of these multiple configurations to obtain a single relay setting for all possible network configurations. In [12], the combination of inequality constraints of all network configurations is converted to interval constraints. Accordingly, the number of constraints has been significantly reduced. However, both algorithms [11,12] are applied only for the

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grid-connected operation mode, and only one type of DG units (synchronous-based distributed generation (SBDG) units) has been taken into consideration. In [13], a hybrid GA-LP method is proposed to solve the protection coordination problem for both radial and meshed distribution systems. Different network configurations are considered, including grid-connected and islanded operation modes, single line outage, and single generator outage. However, only SBDG units are considered, and the overall operating time increases significantly when the constraints of all network configurations are involved in the optimization problem. In [14], the protection coordination problem has been solved using a *meta*-heuristic algorithm for radial distribution systems considering multiple network configurations. When a fault occurs, the network switches reconfigure the network topology to isolate the faulted zone and maximize the load restoration. Accordingly, all constraints of all possible network configurations are combined to obtain a single setting for each relay. However, this algorithm is applied only for radial grid-connected distribution system without including any type of DG. In [15], a k-means clustering method is utilized to classify the multiple network configurations into some clusters, where the number of the multiple setting groups within each relay is equal to the number of clusters. The multiple network configurations result from disconnecting line, generator, and transformer. However, this algorithm is applied only for the grid-connected operation mode, and only SBDG units have been taken into consideration. A communication-based fast adaptive protection coordination schemes have been proposed considering variable DG size and single line outages in [16,17], respectively. In [16], a new constraint reduction-based technique has been developed to accelerate the protection coordination solution. In [17], a novel local search technique has been developed, which removes the coordination constraint violations without affecting the overall tripping time of the relays. However, all the above algorithms [10–17] do not consider the impact of partitioning the ADS on the protective relay performance.

This paper assesses the impact of microgrid partitioning on the performance of the protective relays and their time coordination. In addition, a new inverse time admittance relay (ITAR) is proposed for optimal relay coordination, which can significantly reduce the overall tripping time of the primary and backup relays. The protection coordination problem is formulated as a non-linear programming problem to determine the optimal relay settings for the utilized protective relays. The modified PG&E 69-bus equipped with distributed generation and partitioned into several microgrids is utilized as a test case. Both types of DG units, i.e., SBDG and IBDG units, are considered. Besides, different operation modes are tested including grid-connected, islanded, and partitioned operation modes. The main contributions of the manuscript include:

- The effect of partitioning the ADS on the protection coordination problem is assessed, which is not investigated in the current literature.
- A new ITAR is proposed to minimize the overall operating time of the relays under different network configurations.
- The recorded results emphasize that the ITAR characteristic offers better performance than the ITOCR characteristic with respect to proper relay coordination and total operating time.
- A single optimal relay setting for both islanded and partitioned operation modes can be obtained using the proposed ITAR characteristic, unlike in the case of the standard ITOCR characteristic.

The rest of this paper is organized as follows. The characteristics of both ITOCR and ITAR are introduced in section 2. In section 3, the protection coordination problem formulation is presented. The simulation setup and the analysis of the results are shown in sections 4 and 5, respectively. Finally, section 6 summarizes the paper.

2. Characteristics of standard ITOCR and ITAR

In this paper, two well-known types of protective relays (standard ITOCR and ITAR) are utilized to investigate the effect of network sectionalizing into partitioned microgrids on the optimal relay coordination. However, other tripping characteristics can be used. The tripping characteristics of both relays are introduced in the next subsections.

2.1. Standard inverse time overcurrent relay (ITOCR)

According to IEC standard 60255-151:2009, the operating time of the standard ITOCR characteristic can be obtained using the following tripping characteristic equation [18]:

$$t_{ik} = TDS_i \frac{A}{\left(I_{scik}/I_{pi}\right)^{B} - 1} \tag{1}$$

where t_{ik} is the tripping time of relay i when a fault occurs at fault point k. TDS_iandI_{pi} are the time dial setting and pickup current setting of relay i, respectively. $A$and$B$ are relay constants, and they are equal to 0.14 and 0.02, respectively. I_{scik} is the magnitude of the fault current at relay i when a fault occurs at fault point k. Both TDS_iandI_{pi} are considered as the decision variables of the optimization problem. As the ADS sectionalizing into partitioned microgrids is available option to enhance the control and operation of the distribution network [1], the partition of the ADS leads to a wide range of fault current levels, especially in the presence of both SBDG and IBDG units, which have different fault characteristics. Besides, the new topologies of the partitioned microgrids may be completely different from the network topology before the process of partitioning, and the power flow changes from bidirectional to unidirectional in some feeders. Accordingly, the ITOCRs at one end of these feeders and the directional elements of the ITOCRs at the other end are not essential to be activated. Therefore, from a protection point of view, the influence of partitioning the ADS on the performance of the ITOCRs and their coordination time are important to be considered when the ADS is partitioned.

2.2. Inverse time admittance relay (ITAR)

As reported in [19,20], using only one electrical parameter (current) for feeder protection in ADS may not be quite enough. Fortunately, commercial numerical relays can provide user-defined characteristics [21,22]. One of the user-defined characteristics, which can be developed, is the tripping characteristic of the ITAR. In [23–25], an ITAR has been presented, which is based on a similar concept to distance relays. The tripping characteristic of this ITAR is formulated as follows:

$$t_{il} = \frac{\alpha}{\left(Y_{mil}/Y_t\right)^{\beta} - 1} + \gamma \tag{2}$$

where Y_{mil} is the magnitude of the measured admittance at relay i for a fault occurrence at location l and is given by;

$$Y_{mil} = \frac{|I_{mil}|}{|V_{mil}|} \tag{3}$$

where || denotes the absolute value, I_{mil} and V_{mil} are the measured per-unit fault current and voltage at the location of relay i for a fault occurrence at location l, respectively, Y_t is the magnitude of the per-unit line admittance to be protected and is equal to the reciprocal of the line impedance, and α, β, and γ are constants. The values of these constants are different from that of the standard OCR, and they are determined based on the required protection function of the relay and its location. For example, α, β, and γ are set at 0.0037, 0.08, and 0.005, respectively, in [24]. It is expected that, as the fault point gets close to the relay location, Y_{mil} increases while the tripping time decreases. In contrast, Y_{mil} decreases while the tripping time increases when the fault point

moves further away from the relay location.

The shortcomings of the presented relay characteristic in (2) can be summarized as follows:

(1) The presented characteristic in (2) relies on a fixed pickup value for (Y_t), which limits the possible relay characteristic as well as the relay operating time. To illustrate further, for the standard OCR, the pickup current setting value (I_{pi}) can be optimized and set to a value that minimizes the relay operating time [12,26]. The presented relay characteristic in (2) will operate if Y_{mil} is higher than Y_t. As indicated earlier, the pickup setting of this relay (Y_t) is kept constant irrespective of the different operation modes of the microgrid, and thus constraining the relay operating time.
(2) The values of the IBDGs fault currents are typically within 1.5–2 per-unit [27]. Accordingly, the value of Y_{mil} may not be higher than Y_t for some fault conditions and the relay may be prone to under-reach in these cases.
(3) For the standard OCR, allowing TDS to vary and be optimally selected enhances the relay operation time. However, the presented characteristic in (2) has only one curve corresponding to the selected values of α, β, and γ, and this constrains the relay operating time further.
(4) The tripping characteristic in (1) for a given setting operates for both primary and backup protection. However, the tripping characteristic in (2) provides only one protection function (primary or backup protection) for each selected values of α, β, and γ. In other words, a number of desired protection zones can be generated for different values of Y_t. Accordingly, to provide both primary and backup protection, dual tripping characteristic settings are required; one as primary protection for the specified line and the second as backup protection for the adjacent line connected to the specified line.

To overcome these obstacles, the following tripping characteristic utilizing the magnitudes of the voltage and current in the form of admittance is introduced and can be defined as:

$$t_{ik} = TDS_i \frac{A}{\left(Y_{scik}/Y_{pi}\right)^B - 1} + C \tag{4}$$

where t_{ik} is the tripping time of relay i when a fault occurs at fault point k.TDS_i and Y_{pi} are the time dial setting and the per-unit pickup admittance setting of relay i, respectively. A A, B, and C are relay constants. A and B are equal to 0.14 and 0.02, respectively, as the same as for those in (1), while the selected value of constant C is discussed later. Y_{scik} is the per-unit calculated admittance at relay i when a fault occurs at fault point k. Y_{pi} $\mathrm{Y_{pi}}$ and $\mathrm{Y_{scik}}$ and Y_{scik} are equal to:

$$Y_{pi} = \frac{I_{pi}}{V_{pi}} \& Y_{Scik} = \frac{I_{scik}}{V_{scik}} \tag{5}$$

where V_{pi} and I_{pi} are the per-unit pickup voltage and current of relay i, respectively. V_{scik} and I_{scik} are the magnitudes of the per-unit voltage and fault current at relay i when a fault occurs at fault point k, respectively. Both TDS_i and Y_{pi} in (4) are considered as the decision variables of the optimization problem. The minimum boundary of Y_{pi} can be calculated from (5) based on the minimum boundary of I_{pi} and the upper permissible boundary of V_{pi}. In normal conditions, the maximum permissible voltage drop is assumed equal to 0.1 per-unit [28–30]. Accordingly, the upper permissible boundary of V_{pi} equals 0.9 per-unit. The lower boundary of I_{pi} is determined taking into consideration the maximum permissible overload [12,26]. Accordingly, the lower boundary of I_{pi} is set to 120% of the maximum load current. Based on the selected upper permissible boundary of V_{pi} and lower boundary of I_{pi}, the minimum boundary of Y_{pi} is set at 1.34 per-unit (1.2/0.9 per-unit). To determine the value of C in (4), it takes into consideration that the value of V_{scik} is approximately zero when a fault occurs close to the location of relay i. Accordingly, the operating time of relay i will be equal to zero if the value of C is set at zero in (4). Therefore, the value of C is utilized in (4) to ensure that the operating time of relay i is not equal to zero when a fault occurs close to the location of relay i. As reported in [31], the minimal relay tripping time is typically 16–20 msec. Accordingly, the value of C is assumed equal to 16 msec. It is important to note that the ITAR can be utilized with a directional element similar to the one utilized for the standard directional overcurrent relay (DOCR).

3. Formulation of the relay coordination problem

Practically, the magnitude of the fault current varies with different fault types. The fault current magnitude in the case of ground faults is mainly dependent on the type of the earthing system [32]. The ground fault current in the case of the earthing system earthed through high or low impedance is much lower than that solidly earthed. In most cases, the ground fault current is limited by the neutral ground impedance [33], unlike in the case of the 3-phase faults. Therefore, it is not possible to obtain one relay setting for ground and non-ground faults considering the wide range of fault current values. In practice, the overcurrent relay coordination has two coordination modes; phase mode and ground mode. The phase mode is utilized for non-ground faults (3-phase faults), while the ground mode is utilized for ground faults (1-phase to ground faults) [34,35]. In addition, the overcurrent relay coordination is usually implemented considering bolted ground and non-ground faults (zero fault impedance) since it is considered the worst severe scenario. As the magnitude of the fault current in the case of 1-phase to ground fault is mainly dependent on the system voltage and the neutral ground impedance, the effect of network sectionalizing into partitioned microgrids on the optimal protection coordination is studied in the case of the phase coordination mode in this paper, as it represents the worst scenario. Further, it can be noted that similar results will be obtained in the case of the ground coordination mode.

Determining the optimal settings ($\mathrm{TDS_i}$ and $\mathrm{I_{pi}}$) for the ITOCRs and (TDS_i and Y_{pi}) for the ITARs is the main objective of developing the protection coordination problem for different scenarios while minimizing the total operating time of all primary relays and their backup relays as well as satisfying all required constraints. In this paper, the total operating time is the sum of the operating times of all primary relays and their backup relays due to bolted 3-phase faults at three different locations on each line in the ADS. Thus, the objective function is defined to minimize the sum of the operating times of all primary relays and their backup relays, as follow:

$$Minimize\ \ TOT = \sum_{i=1}^{N} \sum_{k=1}^{K} \left(t_{ik}^{p} + \sum_{j=1}^{Mi} t_{ik}^{b_j ik} \right) \tag{6}$$

where TOT refers to the total operating time. N and K represent the overall number of relays and fault locations, respectively. The symbols "p" and "b_j" $\mathrm{''p''}$ and $\mathrm{''b_j''}$ refer to the primary relay and its backup relay j, respectively. The symbol "Mi" denotes the sum of all backup relays for primary relay i. t_{ik}^{p} and $t_{ik}^{b_j ik}$ represent the operating time of primary relay i. and its backup relay j when a fault occurs at fault point k, respectively. For each line, 3-phase faults are conducted at three fault points, including near-end (1% of line length), mid-line, and far-end (99% of line length) [36,37].

The optimization target of the relay coordination problem is to minimize the total tripping time for both primary and backup relays, which includes.

For the standard ITOCR characteristic shown in (1), the two variables (TDS_i and I_{pi}) of relay i are the decision variables of the optimization problem. Due to the nonlinearity of the tripping characteristic in (1), the optimization problem is defined as a constrained nonlinear

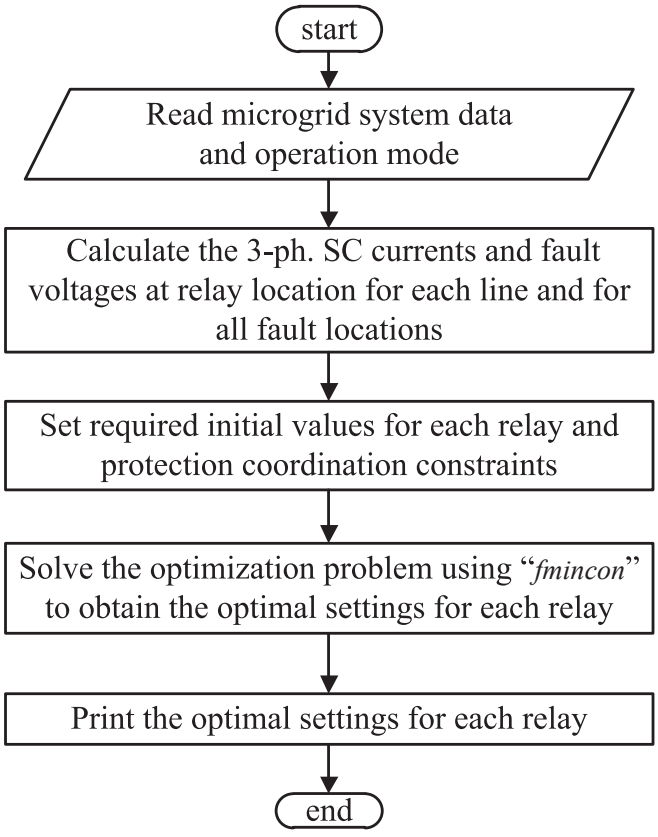


Fig. 1. Required steps to achieve the optimal relay settings.

programming problem to determine the pickup and the time dial settings for each relay. The constraints of the optimization problem imposed on the ITOCR settings can be written as:

$$I_{pi-min} \le I_{pi} \le I_{pi-max} \tag{7}$$

$$t_{ik}^{bjik} - t_{ik}^{p} \ge CTI \forall i, \{k, j\} \tag{8}$$

$$TDS_{i-min} \le TDS_i \le TDS_{i-max} \tag{9}$$

where I_{pi-min} and I_{pi-max} are the minimum and maximum boundaries of the pickup current setting of relay i, respectively, t CTI is the coordination time interval between primary and backup relays, TDS_{i-min} and TDS_{i-max} $\mathrm{TDS_{i-min}}$ and $\mathrm{TDS_{i-max}}$ are the minimum and maximum boundaries of the TDS of relay i, respectively. The minimum boundary of I_{pi} (I_{pi-min}) is set to 120% of the maximum load current. The CTI should be maintained to ensure that the backup relay operates in case primary relay fails to operate. Typical settings of CTI are between 0.2 and 0.5 s, where CTI of 0.3 s is selected in this paper. The other constraint is the range of the TDS settings, which is related to the relay manufacturer. The TDS_{i-min} and TDS_{i-max} $\mathrm{TDS_{i-min}}$ and $\mathrm{TDS_{i-max}}$ are set at 0.05 and 1, respectively [13].

For the ITAR characteristic shown in (4), the two variables (TDS_i and Y_{pi}) of relay i are the decision variables of the optimization problem. Also, due to the nonlinearity of the tripping characteristic in (4), the optimization problem is a constrained nonlinear problem to determine the pickup and the time dial settings for each relay. The constraints of the optimization problem for the ITAR are similar to those of the standard ITOCR except for (7), which is replaced with the following equation:

$$Y_{pi-min} \le Y_{pi} \le Y_{pi-max} \tag{10}$$

where Y_{pi-min} and Y_{pi-max} are the minimum and maximum boundaries of the pickup admittance setting of relay i, respectively. The minimum boundary of Y_{pi} is set at 1.34 per-unit, as explained previously in subsection 2.2. The CTI is set at 0.3 s, same as in the case of ITOCR. In addition, TDS_{i-min} and TDS_{i-max}, are set at 0.05 and 1, respectively, similar to the case of ITOCR. The steps are provided in Fig. 1 to achieve the optimal relay settings.

4. Simulation setup

The PG&E 69-bus ADS [3] is utilized to investigate the impact of partitioning the ADS on the performance of the ITOCR and ITAR

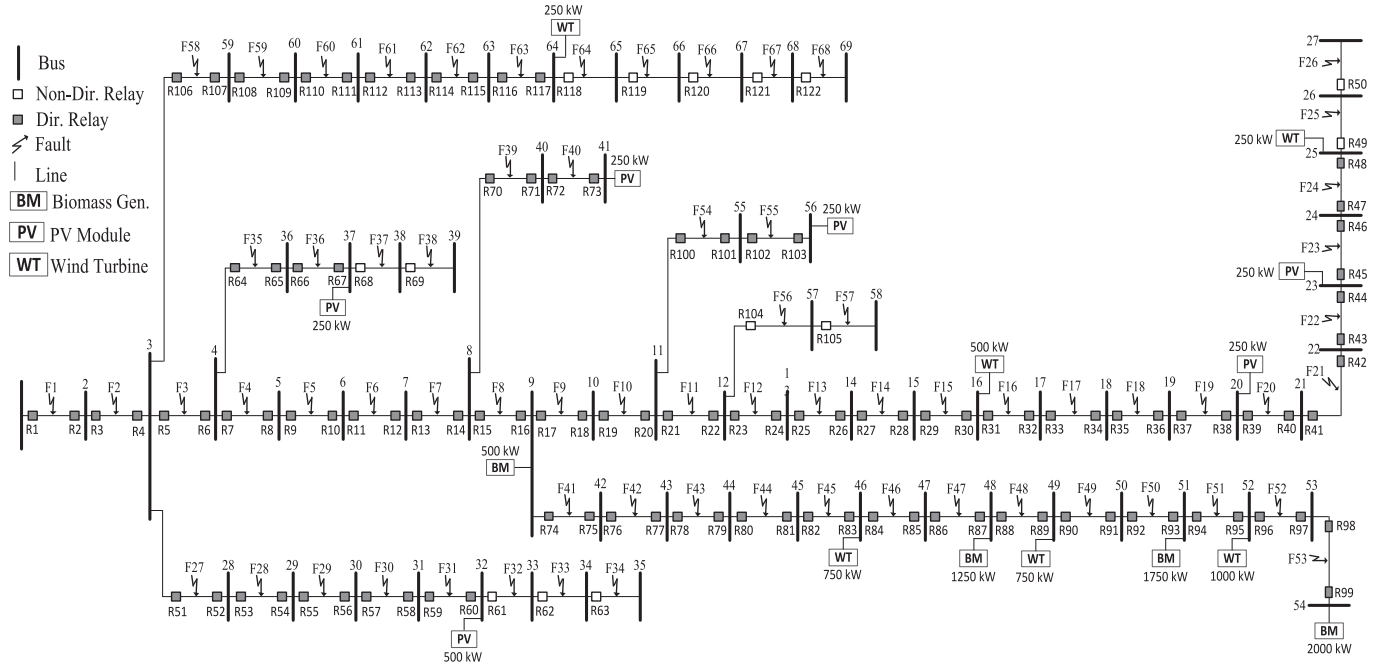


Fig. 2. PG&E 69-bus active distribution system (grid-connected mode).

Table 1
Simulated test cases.

Case	Considered topology
Case-1	Grid-connected operation mode
Case-2	Islanded operation mode
Case-3	Partitioned operation mode
Case-4	Islanded operation mode + Partitioned operation mode

characteristics. The lines and loads data of this system are given in [28,29]. The base apparent power and voltage are 100 MVA and 12.66 kV, respectively. In [3], the optimal size and location of both types of DG units have been determined, as shown in Fig. 2, where the SBDGs units include biomass (BM) generators, and the IBDG units include wind turbines (WTs) and photovoltaic (PV) modules. The total rated powers of the BM, WT, and PV units are equal to 550 kW, 350 kW, and 175 kW, respectively, and they are distributed, as shown in Fig. 2 [3]. In general, the fault current contribution value of each IBDG unit is usually within 1.5–2 per-unit [27], and it is assumed to be 1.5 per-unit in this study. Different cases are conducted in the next sections, and a comprehensive study is provided, including grid-connected, islanded, and partitioned operation modes.

The required computations are executed using MATLAB R2018b on a 2.2 GHz Intel® Core™ i7-8750H CPU with 16 GB of RAM. The optimization problem is solved utilizing "fmincon" MATLAB function. "*fmincon*" function has the capability to find the minimum of a constrained nonlinear multivariable function [38], where different techniques can be utilized to solve the optimization problem [38]. In this paper, the sequential quadratic programming (SQP) technique is utilized to solve the optimization problem in (6). Details of the SQP technique are available in [38].

5. Analysis of results

In Table 1, four different cases are conducted. Firstly, the optimization problem is solved for case-1 (grid-connected operation mode) and case-2 (islanded operation mode). For case-2, the effect of partitioning the ADS on the relay operating time and the protection coordination between primary and backup relays are verified for both ITOCR and ITAR characteristics, where the same relay settings obtained for islanded operation mode are utilized after ADS partitioning. In case-3, the optimization problem is solved separately for each partitioned microgrid for both ITOCR and ITAR characteristics. Finally, in case-4, the optimization problem is solved using the ITAR characteristic taken into account both islanded and partitioned operation modes.

As the ADS experiences various changes due to sectionalizing the ADS into partitioned microgrids, it is worth noting that the optimal settings of either the standard ITOCR or the ITAR tripping characteristics are required to be updated according to the new partitioned microgrids. Therefore, the network sectionalizing the ADS into partitioned microgrids is required to be detected, and hence, there is a need for the availability of the communication system to update the settings of the relays based on the new partitioned microgrids in order to achieve minimal relay operating times. Modern numerical relays have the communication capabilities and facilities that allow users to store several setting groups on the relays. Each setting group corresponding to a given network topology is obtained off-line and stored on the relays as an independent setting group. Accordingly, the appropriate setting group is activated via communication ports according to the corresponding topology. The topology changes can be detected via the control center, which updates the relays with the new topology to activate the appropriate setting group.

5.1. Case-1: Grid-connected operation mode

In Fig. 2, the PG&E 69-bus ADS is equipped with 122 relays (14 non-directional relays and 108 directional relays), named as (*R1* to *R122*). For each line, 3-phase faults are conducted at three fault points, including near-end, mid-line, and far-end. The fault points are named as (*F1* to *F68*). The results of the optimal relay settings for both ITOCR and ITAR tripping characteristics are given in Table 2. As it is difficult to show the results for all relays (122 relays) due to space limitations, the settings of some relays (*R1* to *R30*) are only presented in Table 2. The optimal settings of each ITOCR (I_p and *TDS*) are given in the second and third columns, respectively, while the optimal settings (Y_p and *TDS*) of each ITAR are given in the last two columns, respectively. For instance, the relays (R1 and R3) may have approximately the same pickup value in the case of the ITOCR as well as in the case of R2 and R4. However, as given in IEC equation (1), the TDS has also a significant impact on the operating time of the relay. In the proposed optimization problem, not only the pickup current value is optimized, but also the TDS is optimized to minimize the operating time. Therefore, although the pickup value of R3 is about twice that of R1, TDS of R3 is less than half that of R1. The same is applicable in the case of R2 and R4. Thus, the operating times of both R1 and R3 are close to each other as well as in the case of R2 and R4. The obtained optimal settings ensure proper coordination for both ITOCR and ITAR at all fault points. Besides, the obtained TOTs are 1699.8 and 436.4 sec for both ITOCR and ITAR, respectively, considering all fault points. It can be noted that the ITAR characteristic demonstrates better results compared with ITOCR characteristic for the grid-connected operation mode.

Table 2
Case-1: Optimal settings of ITOCR and ITAR.

Relay No.	ITOCR		ITAR	
	***Ip* (p.u.)**	TDS ***TDS* (sec)**	***Yp* (p.u.)**	***TDS* (sec)**
R1	3.4790	0.3030	50.000	0.6818
R2	0.0038	0.0500	0.0042	0.0500
R3	7.3711	0.0735	21.786	0.6221
R4	0.0520	0.1239	50.000	0.2612
R5	7.7792	0.0500	50.000	0.3534
R6	0.4041	0.0500	10.583	0.5262
R7	7.2762	0.0500	1.4878	0.3094
R8	0.4154	0.0500	0.9869	0.7947
R9	6.5762	0.0500	12.016	0.0613
R10	0.4268	0.0500	27.002	0.0500
R11	6.0757	0.0500	20.011	0.0500
R12	0.4362	0.0500	14.524	0.0500
R13	5.8298	0.0500	31.847	0.1258
R14	0.4444	0.0500	12.977	0.1284
R15	1.5876	0.3054	6.1878	0.2351
R16	0.4331	0.0627	5.1548	0.2856
R17	4.2484	0.0501	9.6255	0.0500
R18	0.0178	0.2017	0.1714	0.4527
R19	2.7961	0.0883	6.9675	0.1549
R20	0.0253	0.0500	3.2309	0.2865
R21	2.8192	0.0590	3.4937	0.0996
R22	0.0042	0.9194	0.0118	1.0000
R23	2.4925	0.0500	3.7247	0.0735
R24	0.0042	1.0000	0.0300	1.0000
R25	2.1404	0.0518	4.3888	0.0624
R26	0.0216	0.0500	3.7688	0.1712
R27	1.9867	0.0500	7.1590	0.0500
R28	0.0217	0.0500	5.3555	0.0737
R29	1.9187	0.0500	13.244	0.0925
R30	0.0218	0.0500	2.2514	0.1996

5.2. Case-2: Islanded operation mode

In islanded operation mode, the total number of relays is decreased to 118 relays (14 non-directional relays and 104 directional relays), named as (*R1* to *R118*), as shown in Fig. 3. The fault points are named as (*F1* to *F66*). In addition, load shedding is implemented to balance load and generation. As an example, the operating time of each primary relay and its backup relays (*R102* up to *R118*) due to a 3-phase fault at the mid

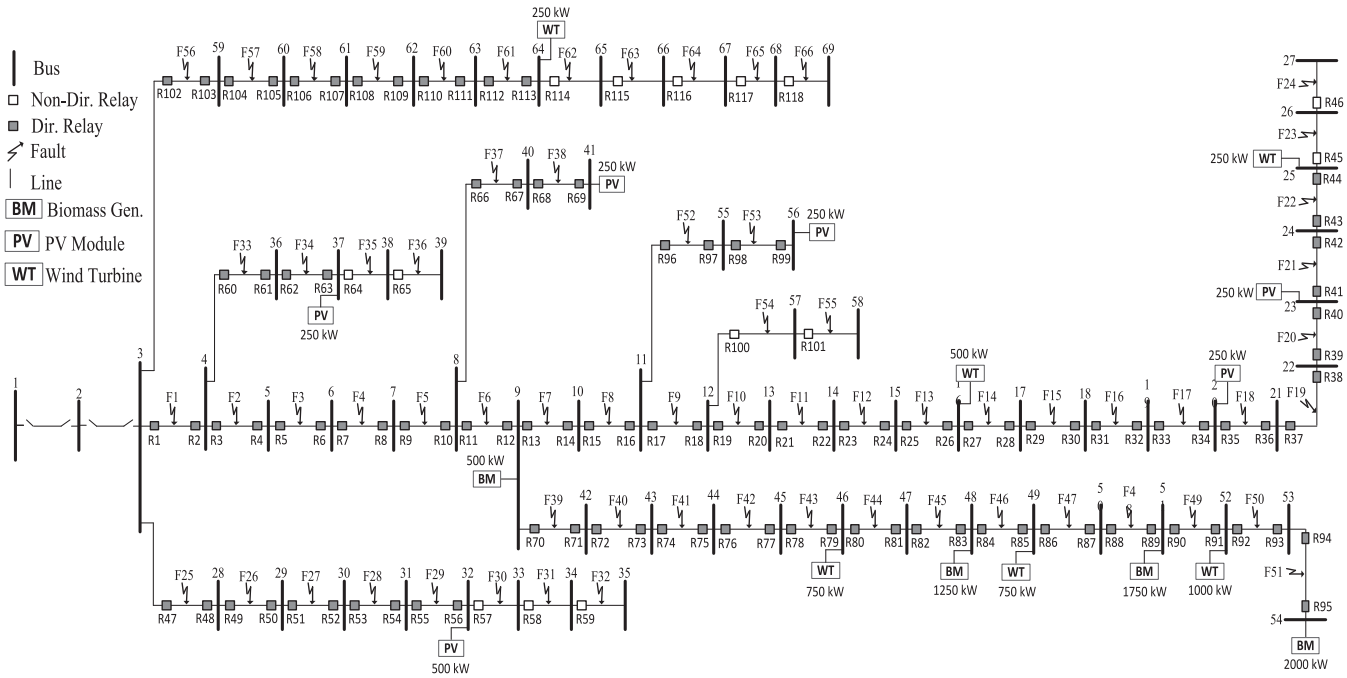


Fig. 3. PG&E 69-bus active distribution system (islanded operation mode).

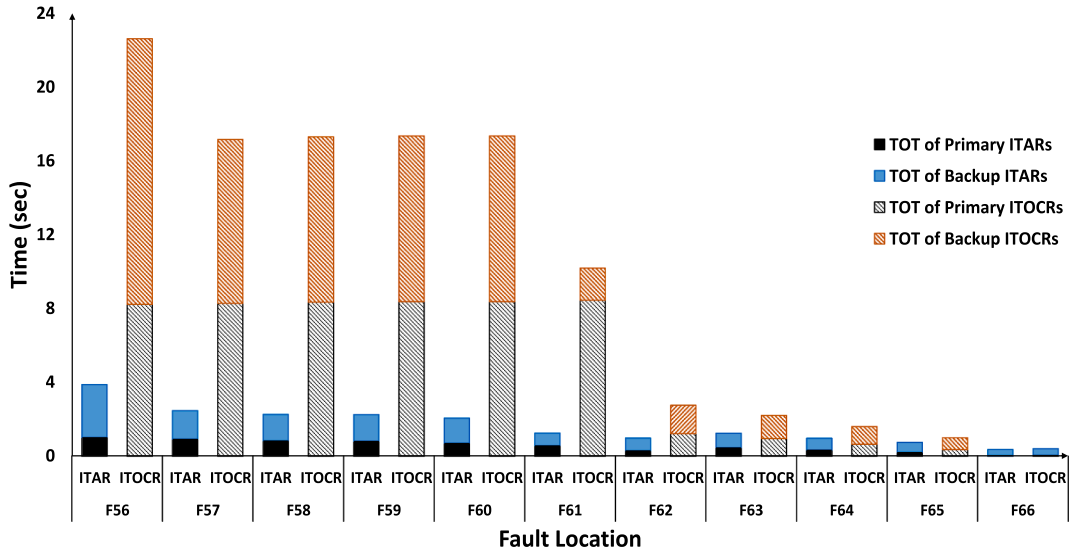


Fig. 4. Case-2: Optimal operating time for 3-phase faults at mid of each line for both ITOCR and ITAR.

of each line are presented in Fig. 4 for both ITOCR and ITAR. As shown, proper coordination between all primary relays and their backup relays is achieved with a minimum CTI of 0.3 sec. In addition, the tripping times of all primary and backup ITARs are lower than those of ITOCR. Furthermore, the recorded TOT for both ITOCR and ITAR are 2561.1 and 376.3 sec, respectively, considering all fault points. Accordingly, the TOT for the ITAR is reduced by 85.3% when compared to ITOCR.

5.3. Case-3: Partitioned operation mode

According to [3], the PG&E 69-bus ADS is sectionalized into partitioned microgrids. The optimum number of partitioned microgrids is equal to eight microgrids based on minimizing the total annual transferred energy between the partitioned microgrids, as shown in Fig. 5. In addition, load shedding is implemented for each partitioned microgrid to balance load and generation. Accordingly, the cut set lines are eight lines and highlighted with red color in Fig. 5. As a result of partitioning the ADS, the power flow in some lines in the partitioned microgrids becomes unidirectional instead of bidirectional in one islanded microgrid. In addition, the two lines (18–19 and 46–47) are disconnected because no loads are connected to buses 19 and 47, respectively. Accordingly, some relays are not required to be activated after network partitioning, and the directional elements of some relays are not essential to be activated due to the unidirectional power flow. The total number of activated relays becomes 81 relays instead of 118 for islanded microgrid, and the number of relays, which their directional elements are not essential to be activated, is equal to 19 relays.

To investigate the influence of network partitioning on the relay coordination, 3-phase faults are conducted at three fault points on each line, as mentioned earlier, in the partitioned microgrids (57 lines). The

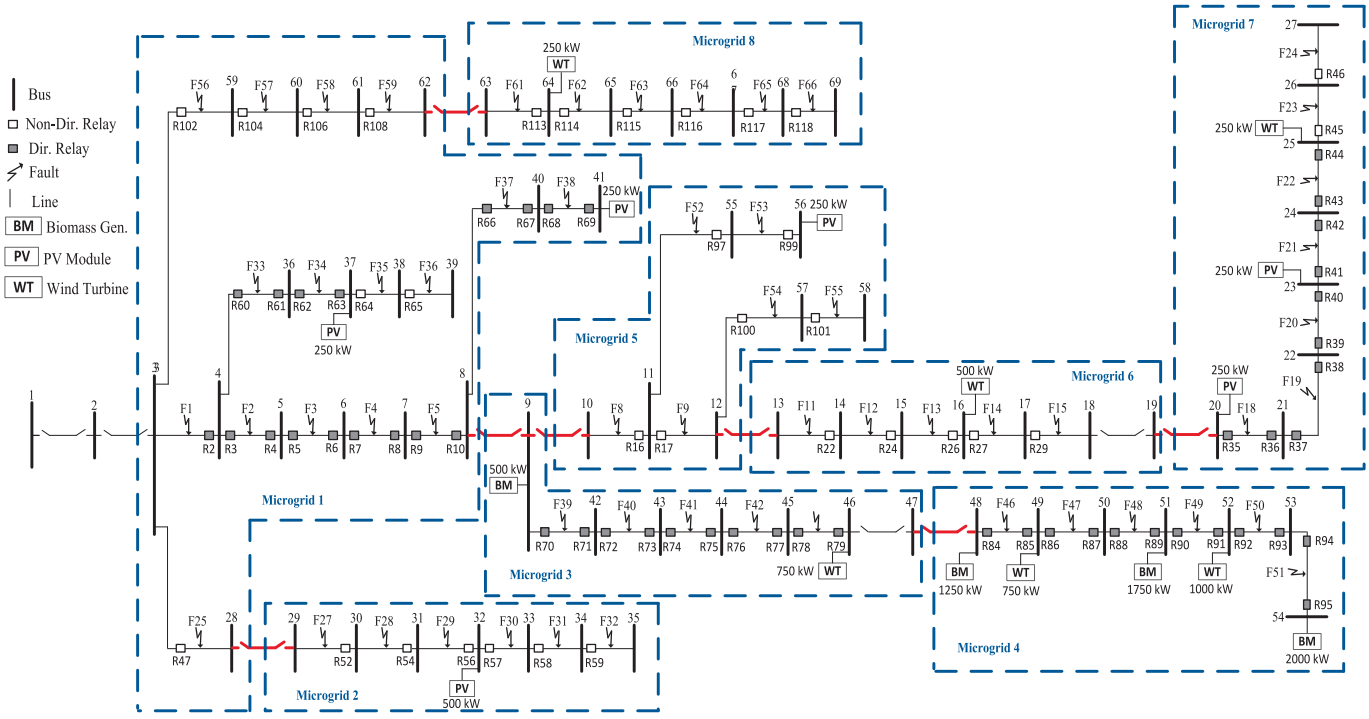


Fig. 5. PG&E 69-bus active distribution system (partitioned operation mode) [3].

Table 3
Number of protection violations for Each Partitioned Microgrid.

Microgrid	Max. Expected No. of PrVs	No of PrVs	
		ITOCR	ITAR
1	210	171	5
2	42	18	0
3	78	78	0
4	102	27	0
5	45	24	3
6	33	33	6
7	96	54	0
8	39	36	0
Total	**645**	**441**	**14**
Total Percentage of PrVs		**68.4%**	**2.2%**

Table 4
Types of protection violations for Each Partitioned Microgrid.

Microgrid	No of PrVs		Type of PrVs					
	ITOCR	ITAR	ITOCR			ITAR		
			1	2	3	1	2	3
1	171	1	54	54	63	0	0	5
2	18	0	6	6	6	0	0	0
3	78	0	30	24	24	0	0	0
4	27	0	9	9	9	0	0	0
5	24	3	9	3	12	0	0	3
6	33	6	15	9	9	0	0	6
7	54	0	18	18	18	0	0	0
8	36	0	12	12	12	0	0	0
Total	**420**	**10**	**153**	**135**	**153**	**0**	**0**	**14**

optimal relay settings for islanded microgrid are used for the partitioned microgrids without any update to these settings to determine the number of protection violations (PrVs), which occur due to network partitioning. Three types of PrVs are considered as follows:

1) ***Type-1:*** Primary relay does not operate.

Table 5
Details of protection violations for ITAR.

Microgrid	Fault Location		Operating time	
			Primary	Backup
1	***F1***	Near-end	**R2**: 0.27	**R61**: 0.55
		Mid-line	**R2**: 0.46	**R61**: 0.57
		Far-end	**R2**: 0.53	**R61**: 0.58
	F33	Near-end	**R3**: 0.37	**R61**: 0.65
		Mid-line	**R3**: 0.32	**R61**: 0.61
5	***F8***	Mid-line	**R16**: 0.39	**R97**: 0.66
		Far-end	**R16**: 0.40	**R97**: 0.60
	F9	Far-end	**R17**: 0.33	**R97**: 0.58
6	***F12***	Near-end	**R24**: 0.47	**R26**: 0.70
		Mid-line	**R24**: 0.28	**R26**: 0.53
		Far-end	**R24**: 0.09	**R26**: 0.22
	F13	Near-end	**R29**: 0.72	**R27**: 0.64
		Mid-line	**R29**: 0.72	**R27**: 0.65
		Far-end	**R29**: 0.72	**R27**: 0.65

2) ***Type-2:*** Backup relay does not operate.
3) ***Type-3:*** CTI between primary and backup relays is less than 0.3 sec, or the backup relay tripping time is less than that for the primary relay.

The number of PrVs is given in Table 3 for each partitioned microgrid. The maximum expected number of PrVs is shown in the second column for each partitioned microgrid, and the number of PrVs is given in the last two columns for both ITOCR and ITAR. As observed, the total percentage of PrVs for the ITAR (2.2%) is much less than that for the ITOCR (68.4%). To further analyze these results, Table 4 presents the numbers of PrVs based on the type of PrV. For the ITOCR, the PrVs occur in all partitioned microgrids. Most of the PrVs cases occur because the primary or the backup relays (type-1 and type-2) do not operate, which affect negatively on the CTI, and increase the number of type-3 PrVs. The ITOCRs do not operate because the levels of fault currents are less than the values of the pickup currents for some fault cases due to the influence of network partitioning. For the ITAR, the fourteen cases of PrVs are due to type-3 PrV. The PrVs occur only in three partitioned

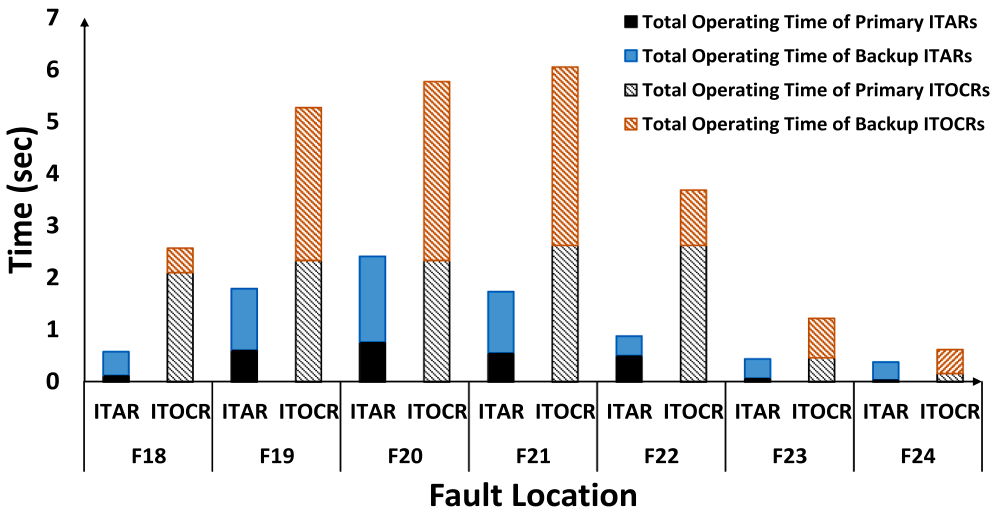


Fig. 6. Case-3, microgrid 7: optimal operating time (sec) for a 3-phase fault at near-end of each line.

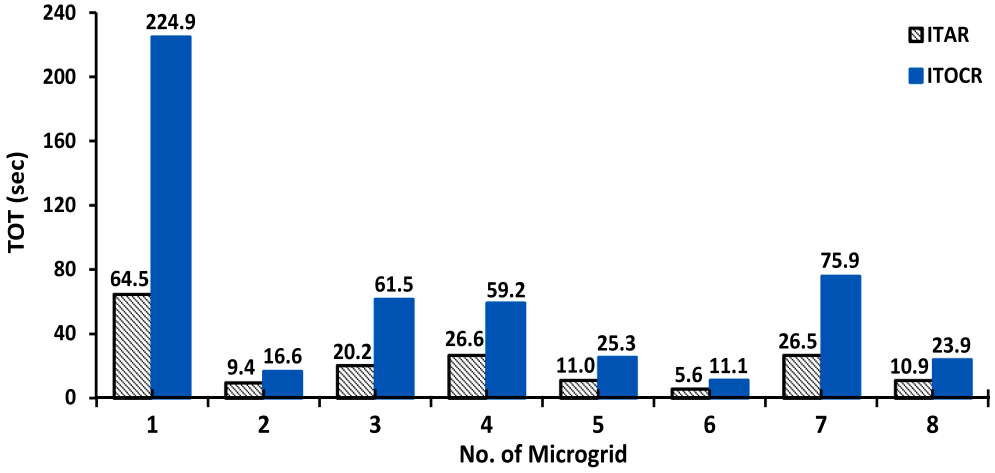


Fig. 7. TOT of ITOCR and ITAR for each partitioned microgrid.

microgrids, which are named as 1, 5, and 6. Unlike the ITOCRs, although the levels of fault currents decrease due to the impact of network partitioning, the values of the admittances also decrease because the magnitudes of the voltages decrease significantly when a fault occurs in either islanded operation mode or partitioned operation mode. In addition, Table 5 displays the tripping times of both primary and backup relays for these fourteen cases. In eleven cases, the primary relays operate before the backup relays with a CTI less than 0.3 sec. In the other three cases, the backup relays operate before the primary relays. However, it can be concluded that the ITAR is still capable of isolating the faulted zone even after network partitioning.

The optimal relay coordination problem can be solved separately for each partitioned microgrid. For each line, 3-phase faults are conducted at the pre-specified three fault points. As an example, the operating time of each primary relay and its backup relays (*R35* up to *R46*) for microgrid No. 7 due to a 3-phase fault at the near-end of each line is presented in Fig. 6 for both ITOCR and ITAR. As shown, proper relay coordination between all primary relays and their backup relays is achieved with a minimum CTI of 0.3 sec. In addition, the tripping times of all ITARs are lower than those of ITOCRs. Further, the TOT of each partitioned microgrid is presented in Fig. 7 for both ITOCR and ITAR. As shown, the TOT in the case of utilizing the ITAR characteristic is significantly less than that in the case of utilizing the ITOCR characteristic. The minimum decrease in the TOT is equal to 5.5 s, while the maximum decrease in the TOT is equal to 160.3 s. It can be concluded that the ITAR achieves a considerable reduction of TOT with respect to ITOCR for all partitioned microgrids.

5.4. Case-4: Single optimal relay settings for islanded and partitioned operation modes

As presented in previous Subsection 5.3, the total percentage of the PrVs for the ITOCR characteristic is equal to 68.4%, which is considered very high. Besides, the fault current levels decrease due to the impact of network partitioning, which are below the minimum pickup current settings in some fault conditions. Therefore, it is not possible to obtain a single optimal relay setting for both islanded and partitioned operation modes using the ITOCR characteristic. Thus, to solve this issue, the protection settings for each operation mode (islanded mode and partition mode) are separately optimized and stored in the relays. Considering the availability of the communication links, the network configuration can be detected, and accordingly, the optimal relay settings can be adaptively modified utilizing the stored protection settings in the relays.

For the proposed ITAR characteristic, the total percentage of the PrVs is equal to 2.2%, which is considered relatively small. In addition, the ITAR is still capable of isolating the faulted zone even when the PG&E 69-bus ADS is sectionalized into partitioned microgrids. Therefore, the

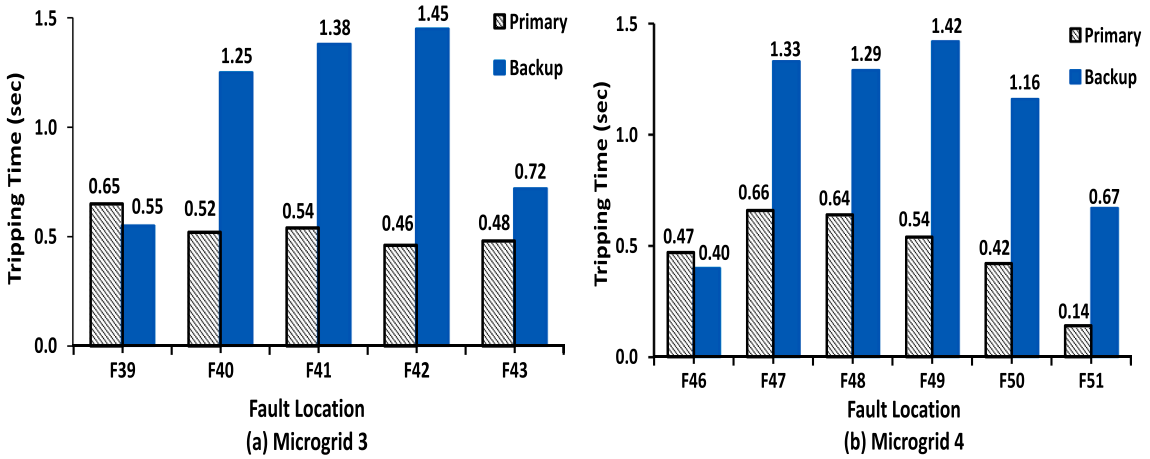


Fig. 8. Case-4: optimal operating time for 3-phase faults at far-end of each line for microgrids 3 & 4.

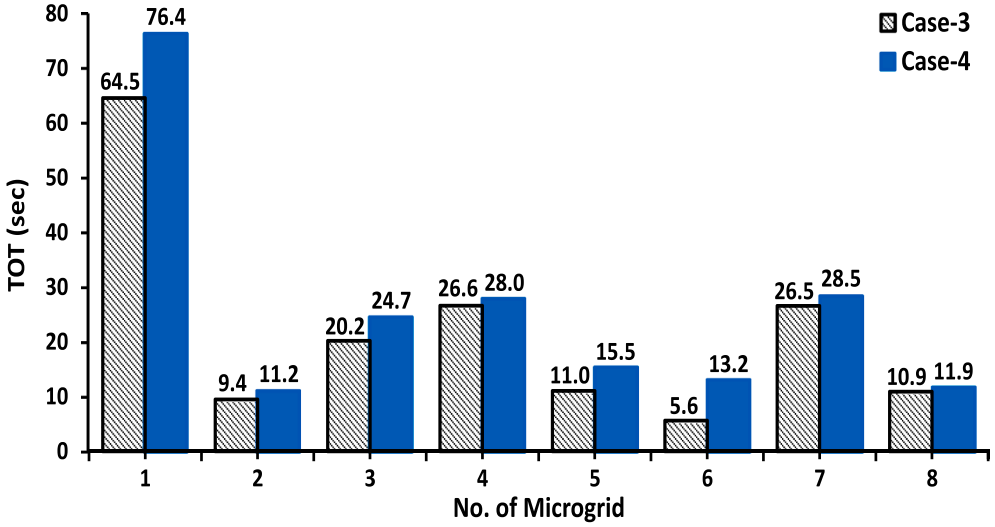


Fig. 9. TOT of partitioned microgrids for case-3 and case-4 utilizing ITAR.

Table 6
Computation time for the tested cases.

Characteristic	Computation time (minutes)			
	Case 1	Case 2	Case 3	Case 4
ITOCR	319.8	384.0	3.0	–
ITAR	1807.7	1265.9	1.6	889.1

optimal relay coordination problem can be solved considering both islanded and partitioned operation modes employing the ITAR characteristic, and a single optimal relay setting can be obtained. To do so, 3-phase faults are conducted on each line at three different fault points. As an example, the operating time of each primary and its backup relays for microgrids No. 3 and No. 4 due to a 3-phase fault at the far-end of each line are presented in Fig. 8. As shown, proper protection coordination between all primary relays and their backup relays are accomplished with a minimum CTI of 0.3 sec. In addition, the tripping times of all ITARs are lower than those of ITOCRs.

In Fig. 9, the obtained optimal adaptive relay settings in case-3 (partitioned operation mode) are compared with those in case-4 (single optimal relay setting for both islanded and partitioned operation modes) with relative to the TOT. As expected, the TOT for the optimal adaptive setting (case-3) is less than those of the single optimal setting (case-4). As shown, the TOT of each partitioned microgrid in the case of utilizing the ITAR characteristic is significantly less than that in the case of utilizing the ITOCR characteristic. The minimum decrease in the TOT is equal to 1.0 s, while the maximum decrease in the TOT is equal to 11.2 s. However, these differences can be neglected. Accordingly, if communication links are not available, a single optimal relay setting for both island and partitioned operation modes can be applied using the ITAR characteristic.

The computation time is provided in Table 6 considering both ITOCR and ITAR characteristics. It is observed that the computation time of case 3 is less than that for the other cases because the optimal settings are obtained separately for each partitioned microgrid. Moreover, the computation time for the ITAR characteristic is higher than that of the ITOCR characteristic because the ITAR characteristic relies on both magnitudes of the voltage and current in the form of admittance, while the ITOCR characteristic relies only on the current magnitude. For case 4, the optimization problem is solved only in the case of the ITAR since it is not possible to obtain a single optimal relay setting for both islanded and partitioned operation modes using the ITOCR characteristic, as explained previously.

5.5. *Simulation results considering high fault resistances*

To investigate the influence of ground faults, the optimal settings are

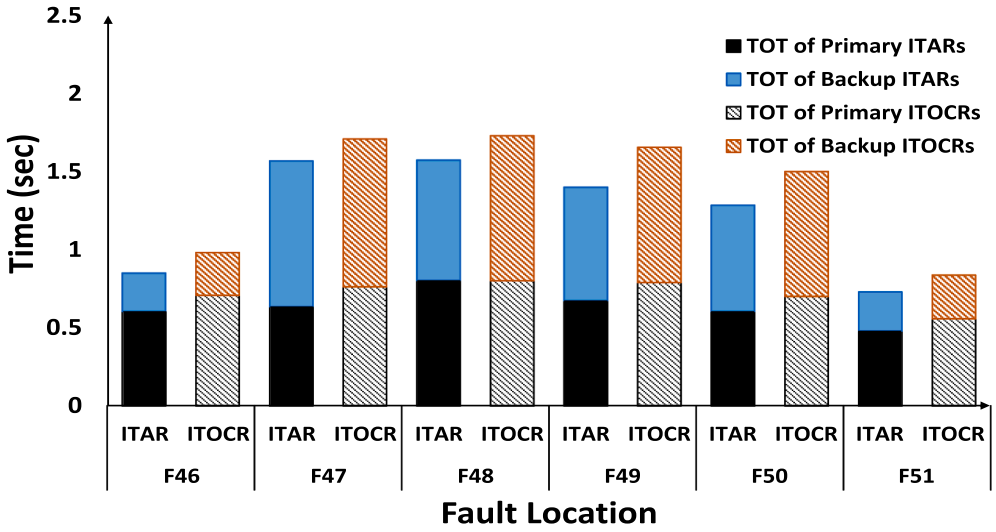


Fig. 10. Case-3: optimal operating time for 1-phase faults at near-end of each line for microgrid 4 with a fault resistance of 50 Ω.

obtained for single-phase to ground faults considering a high fault resistance of 50 Ω. The microgrid 4 in Fig. 3 is considered, and the relay operating times for all lines in microgrid 4 are presented in Fig. 10 for both ITOCR and ITAR tripping characteristics. The minimum CTI for both ITOCR and ITAR tripping characteristics is not less than 0.3 s at all fault locations, and thus both protection schemes do not impose any coordination failure considering single-phase to ground faults. Finally, the TOT of both standard ITOCR and the proposed ITAR tripping characteristics are 8.4 s and 7.4 s, respectively, which highlights the capability of the proposed ITAR tripping characteristic in achieving faster fault isolation than that of the standard ITOCR tripping characteristic.

6. Conclusions

This paper introduces a protection scheme that is based on inverse time admittance relays (ITARs) for active distribution systems (ADSs) with partitioning capabilities. The introduced protection scheme is compared against the standard inverse time overcurrent relay (ITOCR) protection scheme, and the optimization problem is developed as a constrained nonlinear programming problem to determine the optimal relay pickup and time dial settings. Several case studies are conducted on PG&E 69-bus ADS with considerable penetration of both SBDG and IBDG units. Simulation results emphasize that the proposed protection scheme results in a significant reduction in the total relay operating time compared with the standard ITOCR protection scheme considering both grid-connected and islanded operation modes. In addition, for the standard ITOCR characteristic, if the relay settings are not updated after partitioning the PG&E 69-bus ADS, the percentage of protection coordination failure reaches approximately 68.4%. Accordingly, it is not possible to obtain a single optimal relay setting for both islanded and partitioned operation modes using the standard ITOCR characteristic. On the other hand, the percentage of protection coordination failure is approximately 2.2% with the proposed ITAR characteristic. Accordingly, with the this new characteristic, it is possible to obtain a single optimal relay setting considering both islanded and partitioned modes and thus avoiding the need for communication.

CRediT authorship contribution statement

A. Saber: Conceptualization, Methodology, Investigation, Validation, Software, Writing – original draft. **H.H. Zeineldin:** Methodology, Writing – review & editing. **Tarek H.M. El-Fouly:** Conceptualization, Writing – review & editing. **Ahmed Al-Durra:** Investigation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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